Activity 1.3 – 32-bit Barrel Shifter

1. Design Approach:

Designed a 32-bit barrel shifter using Verilog HDL. It supports Logical Shift Left (LSL), Logical Shift Right (LSR) and Arithmetic Shift Right (ASR). A 5-bit input selects the shift amount from 0 to 31.

2. Verification Steps:

- Created a separate testbench.
- Tested left and right logical shifting.
- Tested arithmetic right shifting.
- Applied different shift amounts.
- Checked values with the MSB set.
- Simulated using ModelSim.

3. Results:

The testbench compares actual shifted outputs with expected values and displays PASS/FAIL messages.

4. Issues Faced:

Arithmetic right shift requires a signed operand to preserve the sign bit. Shift amounts must also be checked correctly.

5. Conclusion:

Successfully designed and verified a 32-bit barrel shifter supporting three shifting operations.